Topic 2.3 Why elements react to form compounds

Exercise 2.3A Atoms and ions

1 The outer shell of the atom should be labelled as the highest energy level.

2 Diagram should have a nucleus, and one shell with two electrons.

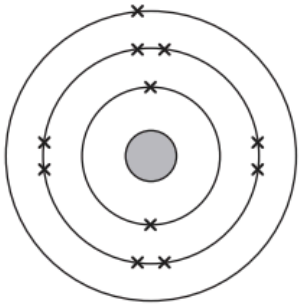
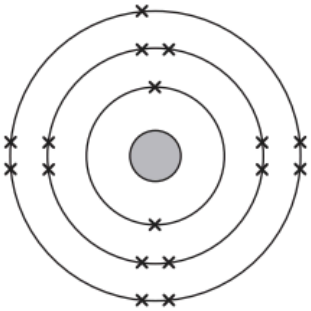
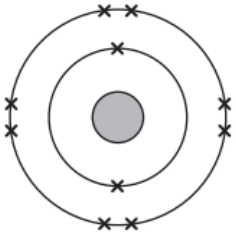
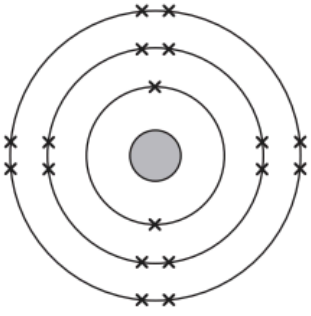
3 Li^+

4 Diagram should have two shells, with electron structure 2,8 and a nucleus.

5 F^-

Exercise 2.3B Why do ions form?

1

sodium: atomic number 11 mass number 23	chlorine: atomic number 17 mass number 35
sodium atom:  The learners must place the electron crosses in the correct shells but they can be anywhere in those shells.	chlorine atom:  The learners must place the electron crosses in the correct shells but they can be anywhere in those shells.
sodium ion:  The learners must place the electron crosses in the correct shells but they can be anywhere in those shells.	chlorine ion:  The learners must place the electron crosses in the correct shells but they can be anywhere in those shells.

2 The electrons are held in place by the electrostatic forces between the protons and the electrons.

3 Ions are formed because the outer shells of the atoms are not full and that makes the atom less stable. The atoms can form full shells by losing or gaining electrons.

Exercise 2.3C Forming ionic compounds

1 a Calcium **atom** diagram should have a nucleus, **four** shells, with electron structure **2,8,8,2**.

Chlorine **atom** diagram should have a nucleus, **three** shells, with electron structure **2,8,7**.

Diagrams must be labelled.

b Calcium **ion** diagram should have a nucleus, **three** shells, with electron structure **2,8,8**.

Chlorine **ion** diagram should have a nucleus, **three** shells, with electron structure **2,8,8**.

Diagrams must be labelled; learners may mark the chlorine ion as negatively charged and the calcium ion as positively charged.

c The calcium atom has two electrons in its highest energy shell. For the calcium atom to become more stable these two electrons must be lost. Chlorine has seven electrons in its highest energy level so only needs one electron to fill this shell. Two atoms of chlorine are needed to use the two electrons lost from the calcium atom so the formula for calcium chloride is $CaCl_2$ because two atoms of chlorine are needed for every one atom of calcium.